

Toxicity Report – Eye Cream Ingredients

This report summarises the available safety and toxicity information for each listed ingredient in the eye-area cream. It is not a medical diagnosis. For personal advice, consult a doctor or dermatologist.

1. Summary Table

Ingredient	Main Role	Known Toxicity / Concerns at Cosmetic Levels	Practical Risk (Normal Adult Skin)
Aqua (Water)	Solvent	No relevant toxicity concerns.	Very low
Glycerin	Humectant (draws water into skin)	Occasional stinging on very damaged skin.	Very low
Cetearyl Alcohol	Fatty alcohol emollient	Rare contact allergy; not a drying alcohol.	Very low
Glyceryl Stearate	Emulsifier, emollient	Very low irritation in tests.	Very low
Ceteareth-20	Emulsifier (ethoxylated)	Considered safe by expert panels; may contain trace 1,4-dioxane if poorly purified; more irritating on broken skin.	Low–moderate (mainly irritation / theoretical contaminant risk).
Niacinamide	Active (vitamin B3)	Good safety data; some people get flushing or stinging, especially with damaged barrier or rosacea.	Low (monitor if sensitive).
Butyrospermum Parkii (Shea) Butter	Occlusive moisturiser	Rare nut-related allergy.	Very low
Squalane	Emollient	Inert and non-sensitising in studies.	Very low
Caprylic/Capric Triglyceride	Emollient	Very low systemic toxicity; can clog pores in some users.	Very low

Palmitoyl Tripeptide-1	Anti-ageing peptide	Expert panels: safe at current cosmetic levels.	Very low
Palmitoyl Tetrapeptide-7	Anti-ageing peptide	Same safety conclusion as above.	Very low
Vaccinium Myrtillus (Blueberry) Fruit Extract	Botanical antioxidant	Usual plant-extract issue: allergy/irritation in a minority.	Low
Butylene Glycol	Solvent / humectant	Generally low toxicity; can cause mild irritation or dermatitis in sensitive people.	Low
Carbomer	Thickener	Large polymer, not absorbed; rare local irritation.	Very low
Polysorbate 20	Surfactant / solubiliser	Considered safe; like other ethoxylated ingredients may contain trace 1,4-dioxane if not well purified; occasional irritation.	Low–moderate (irritation + theoretical contaminant risk).
Sodium Lactate	Humectant, pH buffer	Safe in cosmetics; can sting on broken or very sensitive skin.	Low
Octadecyl di-t-butyl-4-hydroxyhydrocinamate	Antioxidant to stabilise oils	Toxicology data show low acute toxicity; slight skin irritant at higher doses; no strong carcinogenicity signal.	Low (possible irritation/allergy).
Phenoxyethanol	Preservative	Regulators consider it safe up to 1% with good safety margin; main issues are irritation or allergic eczema in some people.	Low–moderate (preservative irritation; avoid if you react to it).

Citric Acid	pH adjuster	Can sting eyes or broken skin at higher levels; low systemic toxicity.	Low
Sodium Benzoate	Preservative	Widely used in food/cosmetics; occasional irritation or allergy, especially on mucous membranes.	Low
Potassium Sorbate	Preservative	Similar to sodium benzoate – rare irritation or allergy.	Low

2. Detailed Notes by Ingredient Group

A. Base and Moisturising Ingredients

These include water, glycerin, cetearyl alcohol, glyceryl stearate, shea butter, squalane, caprylic/capric triglyceride, sodium lactate and citric acid. They mainly act as solvents, humectants and emollients. Toxicology data show very low systemic toxicity. The most common real-world issues are mild stinging on already-irritated skin or rare contact allergy (for example to cetearyl alcohol or shea butter).

B. Actives and Peptides

Niacinamide is the active form of vitamin B3 and has a long history of safe cosmetic use. It can cause flushing or stinging in some users, especially if the skin barrier is compromised or if there is rosacea. Palmitoyl tripeptide-1 and palmitoyl tetrapeptide-7 are signalling peptides used at very low levels; expert safety panels have concluded that they are safe in current cosmetic uses and concentrations. The main theoretical risk is simple allergy, which appears to be rare.

C. Solvents, Surfactants and Thickeners

Butylene glycol, polysorbate 20, cetareth-20 and carbomer help dissolve other ingredients, mix oil and water and adjust texture. Overall toxicity is low, but these are more likely than the basic oils to cause local irritation, especially near the eyes or on broken skin. Polysorbate 20 and cetareth-20 are ethoxylated surfactants. If manufacturing controls are poor, they can contain trace amounts of 1,4-dioxane, a by-product of the ethoxylation process. Modern cosmetic standards require these impurities to be minimised, and when properly purified the finished products are regarded as safe by regulatory and expert bodies.

D. Preservatives and Antioxidant

Phenoxyethanol, sodium benzoate and potassium sorbate prevent microbial growth. Regulatory reviews, including those in the European Union, consider phenoxyethanol safe at up to 1% in cosmetic products with an adequate margin of safety. At normal topical exposure, systemic toxicity is not expected. The practical concern is local irritation or allergic contact dermatitis in susceptible individuals. Sodium benzoate and potassium sorbate have similarly good systemic safety profiles but can occasionally irritate mucous membranes.

Octadecyl di-t-butyl-4-hydroxyhydrocinnamate is an antioxidant that stabilises the oils in the formula. Toxicological data show low acute toxicity and only slight skin irritation at higher experimental doses. In cosmetics it is used at very low concentrations; the realistic concern is, again, potential irritation or allergy in sensitive users.

E. Overall Assessment

Taken together, the ingredients in this eye-area cream do not include any substances that are recognised as high-toxicity or strongly carcinogenic at cosmetic use levels. The realistic risks are local: stinging, redness or rash, especially in people with sensitive or damaged skin or a history of cosmetic allergies. If such symptoms occur, stop using the product and consult a healthcare professional with this ingredient list.